Integrating Digital Biomarkers with Feedback-Driven Clustering for Optimized CPAP Adherence in OSA Management

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***Abstract*—Obstructive Sleep Apnea (OSA) is a significant disorder of cardiovascular health and well-being, but compliance to continuous positive airway pressure (CPAP) therapy is another problem. The existing way of analyzing CPAP adherence relies mainly on the previous compliance data to offer treatments that will delay intervention and increase the risk of therapy termination. An artificial intelligence framework consisting of such functions as biometric monitoring and patient cluster analysis via machine learning is suggested for the development of an individual CPAP therapy plan. The system modifies the pathway of therapy based on consistent monitoring of heart rate variability (HRV), oxygen desaturation index (ODI), and variability in respiration effort measurements. The proposed real time decision support system bests traditional CPAP monitoring systems due to its capability to do individualized interventions where biometric signals together with the participants' response and observed CPAP usage is used in cultivating better compliance results. By combining CPAP machine data, the system improves patient diagnosis category and plans for intervention. Initial results indicate feasibility on 45,000 French patients for the proactive, precision-based shift of CPAP management from retrospective tracking towards improving long-term success of treatment.**

Keywords— Obstructive Sleep Apnoea (OSA), Continuous Positive Airway Pressure (CPAP), CPAP Adherence, Personalized Care, Biometric Feedback, Digital Biomarkers, Adaptive Interventions, AI-driven Personalization, Risk Assessment, Sleep Therapy Optimization.

I. INTRODUCTION

Obstructive sleep apnoea (OSA), a severe disorder which impacts nearly a billion people in the world is characterised by frequent airway blockages during sleep, oxygen replenishment setbacks and altered sleep pattern's greater chance to develop cardiovascular diseases, metabolic disorders and cognitive impairment respectively[1]. Additionally, these disruptions have been linked to neurological conditions, reinforcing the need for timely diagnosis and intervention[2]. Continuous positive airway pressure (CPAP) remains the leading treatment for OSA[3], but patient retention rates remain unsatisfactory up to 30–50% of patients discontinue therapy due to discomfort, unclear therapeutic benefits, stigma, and insufficient patient support[4]. The combination of psychological influences and insufficient understanding of CPAP advantages further increases non-compliance rates. The current adherence frameworks mostly use tracking data to measure adherence defined as ≥4 hours of CPAP use per night[5]. The retrospective monitoring system detects non-compliance only after clinical deterioration occurs, which increases healthcare costs and worsens patient outcomes[6]. The monitoring of ongoing physiological signs including heart rate variability (HRV), oxygen desaturation index (ODI), and respiratory effort variability (RVO) enables early patient risk assessment and timely intervention by healthcare practitioners[7]. Real-time analysis of HRV and ODI trends facilitates earlier intervention to prevent discontinuation of treatment[8]. Despite advancements in biometric monitoring, existing adherence models remain predominantly static, relying on demographic and device usage patterns rather than dynamic, patient-specific physiological trends[9]. This approach limits personalized care, leading to high dropout rates and suboptimal outcomes. Research proposals based on machine learning clustering seek to classify patients based on their CPAP usage patterns together with their risk profiles while utilizing biometric data from wearable devices[10]. These AI-driven models enable precise patient stratification and the delivery of personalized intervention strategies, which have shown potential to improve adherence rates[11]. However, few doctors and hospitals are using AI in healthcare because they have not solved data integration problems, privacy concerns, or how to validate the results. To begin, we applied K-means clustering for profiling the patients in our previous

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study[11]. Our method involves applying Mean Shift clustering to improve CPAP adherence without the necessity of pre-specified cluster numbers. It also monitors biometric data in real time and adjusts intervention based on the patient's condition[12]. This approach combines many data sources to help patients manage their CPAP therapy in a proactive and supportive way that succeeds in detecting risks and encouraging treatment adherence. This model has been validated using live patient data and it is in use for more than 45,000 patients in France.

II. State of the Art

A. Scope and Depth of Literature Review

A structured literature review over the past three years using Web of Science databases analyzed 492 publications about CPAP adherence, 484 documents regarding treatment discontinuation and 487 studies on barriers to adherence and psychosocial aspects. The combination of digital interventions and behavioral support strategies has proven effective in improving sustained CPAP adherence according to research findings[13]. Research on biomarker-based adherence solutions included 64 articles related to digital biomarkers 51 articles about heart rate variability (HRV) and sleep phase and 123 articles examining CPAP interventions using biomarkers. Wearable devices like smartwatches and oximeters provide continuous patient tracking that enhances adherence results[14]. Research by our group confirmed that implementing patient feedback mechanisms into CPAP adherence algorithms generates superior intervention results while improving patient clustering[11]. Real-time feedback models have maintained their significance as critical tools for improving treatment adherence. The literature review revealed key gaps in developing personalized therapies as well as real-time intervention methods and AI-based adaptable models for CPAP therapy approach. The analysis demonstrates the necessity of dynamic, patient-specific treatment protocols based on continuous biometric feedback.

B. Maintaining the Integrity of the Specifications Current Gaps in CPAP Adherence Management

CPAP is the standard treatment for OSA shows insufficient patient adherence because up to 50% of users discontinue therapy within the first year[15]. Psychological barriers, including perceived stigma, anxiety, and lack of education, contribute to low adherence rates, highlighting the necessity of multi-dimensional, personalized interventions[16]. Our previous work [11] demonstrated that integrating patient feedback with CPAP monitoring significantly improved adherence predictions, allowing for early intervention before full withdrawal from therapy. Professional assessments confirm that early treatment interventions are essential since many patients who experience complications within their initial month end up becoming non-compliant in the long term[17]. AI-driven predictive analytics successfully determines patients at risk before therapy dropout through early prevention strategies[5]. Developing real-time feedback predictive models is critical to detecting and mitigating adherence risks. Sleep-related cardiovascular damage alongside metabolic instability and cognitive impairment can be prevented through these models according to the[18].

C. Biomarkers in CPAP Therapy and Their Potential Impact

Research shows that biometric monitoring brings a revolutionary approach to managing CPAP adherence management. Biomarkers used for treatment evaluation include HRV to measure autonomic nervous system activity together with ODI for hypoxia and airflow variability and respiratory effort for total assessment[19]. Artificial signals derived from biometrics offer immediate physiological evidence so clinicians can modify treatments in real-time because of this functionality[20]. The severity of OSA matches with abnormalities in HRV which indicates autonomic dysfunction serves as a leading risk factor for patients who stop following their CPAP therapy[21]. Our previous study[11]analyzed patient survey integration with CPAP monitoring systems demonstrating that combining patient-reported data and physiological measurements enhances adherence prediction and therapy outcomes. The desaturation trend of the SPO_2 shows persistent hypoxia despite CPAP use, while variability of respiratory effort and airflow metrics supports continuous pressure optimization, reducing mask discomfort and tolerance[15]. These promising biomarkers face limitations in clinical usage because healthcare organizations face challenges regarding patient privacy plus difficulties standardizing their testing procedures[7]. Real-time monitoring tools including smartwatches and CPAP integrated sensors with oximeters show effectiveness in therapy improvement but remain underutilized in mainstream distribution[22].

D. Challenges in Clustering and Risk Quantification in Current Frameworks

The primary concern with CPAP therapy lies in the lack of AI tools that can help assign patients into risk categories. Frequently, adherence models for patients are built based on their usage duration and basic demographics, ignoring how physiologically different they may be[15]. While AI models are now using machine learning, the challenge of interoperable data prevents them from being used in medicine[23]. An earlier work by our team[11] applied clustering-based models on CPAP adherence, illustrating the effectiveness of machine learning in intervention desgining. The current approach is expanded through the implementation of automatic real-time patient biometric monitoring. AI-based clustering detects early warning signs of poor adherence and heightened cardiovascular risk[19]. However, three major challenges remain: standardizing data collection, achieving device interoperability, and establishing clinical validation[20]. The advancement of AI technologies has not yet bridged the gap to standardized adherence prediction, which remains an obstacle to widespread clinical adoption[17].

E. Need for a Proactive and Personalized CPAP Model

We start with our earlier framework for feedback-based clustering[11], The model relies on digital signs and real-time collateral to tailor the experiences. The system updates the risk profiles of patients by adapting its approach, allowing for faster intervention and better care[22]. Frequency, spO2 desaturation, and respiration are measured on an ongoing basis to identify any possible therapy stoppage[24]. A personalized approach to CPAP treatment, including managing mask retention times, will likely depend on AI to improve patients' adherence [25]. By using AI and monitoring biometric data, the proactive system maximizes continuation

of therapy, lowers intervals of discontinuation, and results in top treatments.

III. METHODS

The purpose of this study is to introduce a framework that uses AI, biometric data, machine learning, patient grouping, flexible strategies, and ongoing learning for CPAP therapy. The system replaces fixed compliance tracking with a newer, dynamic process that boosts compliance and outcomes. Our previous study indicated that personalized interventions work best when patients' feedback is received and integrated within their CPAP adherence system[11]. This indicates that we have to always adapt and use feedback.

A. Patient Selection and Initial Clustering

The framework begins with a thorough stratification process using both retrospective and real-time data. Unsupervised machine learning techniques help classify patients into different adherence profiles through k-means for future risk prediction[11]. Clusters include multidimensional datasets consisting of:

- **Demographic Data:** Major factors such as age, gender, BMI. Studies have highlighted the importance of including socio-economic and psychosocial factors to improve the accuracy of clustering [26].
- **Physiological Indicators:** CPAP device data, including usage hours, mask leakage rates, residual AHI levels, pressure settings. Emphasis is also placed on the dynamic correlation between leakage patterns and adherence outcomes[9].
- **Behavioural Insights:** Therapy interruptions, mask type preferences, and responses to monitoring surveys. Wearable devices tracking daily activity and lifestyle enhance predictive accuracy [27].
- **Medical Survey Responses:** Medical evaluations using validated surveys PSQI, ISI, ESS, FSS, SF-36, and BDI offer empirical data about sleep quality together with fatigue levels and psychological health[11].

The initial group identified through clustering as high risk will receive priority for rapid intervention according to the first iteration is prioritized for immediate intervention Fig.1. Ambiguous clusters are further assessed with biometric and behavioral measures including HRV, RVO, mood, and activity level data[22]. Our research [11] showed that CPAP users can benefit from cluster analysis of adherence data which supports improving treatment through patient-specific intervention profiling.

B. Risk Evaluation and Stratification

Following the initial clustering, the process of evaluating risk iteratively, as illustrated by the second iteration in Fig.1. At this stage, the program checks how well a patient is doing with CPAP and if medical or behavioral changes are required. The markers to consider are how well the patient sticks to CPAP, device leakage, the remaining AHI, the pressure set by the device, and the survey results. Monitoring real-time physical data on a wearable strengthens risk assessment by revealing even minor changes that people often hide during interviews[23]. Clinical risks (like heart disease, obesity, and smoking) and physical data (like HRV and oxygen saturation) are routinely studied. Risk factors are constantly redefined using machine learning so that the recommendations are more accurate[28]. This score predicts the possibility of not continuing treatment, giving way for early support. People are given a preliminary risk rating, and their levels are reviewed and updated as more information is available. Regular checking and survey data help ensure the strategies fit each person best.

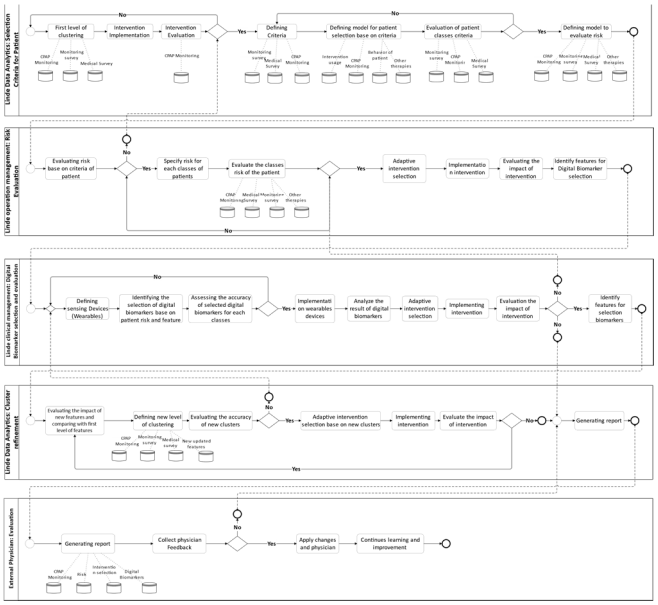

Fig. 1. Adaptive Digital Biomarker-Integrated Clustering Framework for Personalized CPAP Adherence Management.

C. Digital Biomarker Selection and Evaluation

In this stage, the framework focuses on selecting appropriate digital biomarkers, emphasizing real-time physiological monitoring, as illustrated in the third iteration of Fig.1. Biomarkers such as HRV, ODI, and RVO are prioritized due to their relevance in predicting adherence behaviour. Recent studies highlight the value of combining biometric data from multiple sources, such as smart watches and oximeters, to improve predictive accuracy[29]. The choice of the wearables, including sensors integrated in the CPAP, smart watches and oximeters, is based on the risk profile of the patient and the variability of biometrics. Comparative studies between these biomarkers and CPAP data are done frequently to verify their usefulness and accuracy. In cases where biomarkers have low predictive power, new ones are introduced to keep clinical value. Feedback algorithms help choose biomarkers based on the patient's current status, improving accuracy[9]. The effectiveness of using the chosen devices will be assessed by analysing the results from the integration to see if real-time monitoring and adherence are effectively improved. During this process, it's ensured that using biomarkers is both solid and adapted to the needs of every patient.

D. Adaptive Intervention Implementation

The adaptive intervention module reacts to patient risk profile changes Fig.1.The system modifies intervention approaches after receiving steady flow of information from CPAP monitoring devices as well as doctor-provided biometric data and medical surveys. The intervention involves dynamic CPAP pressure control together with individualized behavior change programs that utilize automatic feedback systems[22]. For patients identified as high risk, targeted interventions include automatic adjustment of CPAP,

behavioural coaching, lifestyle modification recommendations and individualized reminders of adherence. Our research demonstrated that feedback-driven CPAP interventions improve patient adherence [11]. The systematic approach includes repeated feedback cycles which ensure the interventions adapt effectively to patient system modifications. The method reduces patient interruption while making the intervention more beneficial for the patient. The continuous monitoring process helps healthcare staff identify intervention impacts quickly and adjust treatment methods continuously to achieve maximum results. The process operates through feedback loops which validate intervention effectiveness by tracking real-time patient data and direct patient feedback to maintain personalized responses for individual patient dynamics.

E. Cluster Refinement and Continuous Learning

Cluster refinement iteratively improves cluster definitions by assessing new features against prior groupings, as depicted in the fourth iteration of Fig.1. Data sources such as CPAP monitoring, up-to-date medical data and new biometric elements are integrated to define new clustering levels. The accuracy of these associations is rigorously assessed, and treatment pathways are refined based on the available patient data. If new clusters demonstrate improved predictive capabilities, they shall be accepted for subsequent stages of intervention. This approach ensures that interventions remain in line with the behaviour of patients in real time and with clinical needs[22]. This continuous learning ensures the system adapts to patient behavior changes and clinical needs. The iterative learning mechanism improves the accuracy of the grouping, allowing the system to identify high-risk patients more accurately and to recommend appropriate interventions.

F. Physician Evaluation and Feedback Loop

The final step produces integrated reports for physician review which can be seen in the fifth iteration of Fig.1. Reports include CPAP data, risk assessments, biometric trends, and intervention results. Real-time dashboards aid clinicians in treatment decision-making[22]. Physicians' feedback is incorporated to further refine and personalize intervention strategies. The implemented changes should go into effect straight away to maximize treatment success. The feedback loop supports an ongoing system enhancement through continuous learning which keeps the framework accurate in clinical practice and responsive to changing patient requirements. Adaptive algorithms use physicians' information to merge expert knowledge with automated technology for intervention accuracy improvement[23]. The combined AI-clinician approach enhances adherence by shifting from static data monitoring to proactive, personalized therapy, as it will not only rely on algorithms but also include expert clinical opinion, improving overall treatment personalization and patient adherence outcomes.

IV. IMPLEMENTATION

Research continues from our initial study[11], built upon the study that applied K-means clustering to classify OSA patients eligible for CPAP treatment, and we discovered drawbacks regarding cluster flexibility and responsiveness to patient diversity. K-means requires predefining the number of clusters, which may not capture the complex patterns in patient adherence behavior. To solve these issues, we decided to use Mean Shift clustering, a type of algorithm that detects clusters in data based on how densely they appear without predetermining the number of clusters. This study suggests a method that iteratively uses feature analysis, regular checkup recommendations, and tailored treatment options. This framework also aims to use digital markers to boost patient adherence and enhance medical results.

A. Data Collection and Preprocessing Framework

Consistent with the approach in our previous research, this study included extensive CPAP monitoring data from Philips, ResMed, Sefam, and Lowenstein devices, along with detailed patient records, risk factors, intervention results, and structured monitoring survey data (28% female, 71% male). Data collection has been strengthened through a few channels, including personal visits, mobile applications, SMS and telephone calls, to ensure a comprehensive coverage of patient behaviour. By integrating multiple data sources, we aimed to reduce bias and improve the reliability of clustering [23]. Multiple pre-processing techniques have been used to ensure data quality and consistency. SimpleImputer has handled missing values by imputation of averages or frequent values. The LabelEncoder was used to encode categorical variables, and the StandardScaler standardised the numeric properties for the single analysis. ColumnTransformer facilitates different preprocesses across datasets, ensuring optimal preparation for clustering analysis. This structured and robust data engineering framework guarantees high quality clustering input and allows for targeted interventions [9].

B. Initial Clustering Iteration:CPAP Data

In the first iteration, we reviewed data about CPAP use, focusing on essential statistics like the Apnea-Hypopnea Index (AHI), LeaksPercentage, and DeviceType. To assess consistency, we relied on patient record dates and measured the UsageDuration. The reason for using the mean shift algorithm was its ability to evaluate how patients' clusters change[30]. By applying Mean Shift clustering on this dataset, it was possible to separate it into nine clusters that showed Silhouette Score and Davies-Bouldin Index values of 0.37 and 0.76, respectively. The care team tailored strategies according to the different adherence profiles in each cluster. For instance, groups with less transgressions were taught adherence strategies, those with significant airflow leakage were provided mask fit assistance, and patients with excessive airway obstruction received interventions specific to apnea. The initial phase focused on defining clinical aspects that could guide the development of digital biomarkers. The original number of nine clusters was kept, but there was now a better fit between the strategies and who needed them. By using an iterative method, the clusters are updated to stay current and relevant for patients[28].

C. Intermediate Clustering Iteration: Integrating Risk Factors and Patient Data

During the second round, more information was added, such as Risk Factor data, Patient Data, and Mask data. Both Risk Factor and Patient Data had fields like 'RiskFactorValue', 'RiskFactorDescription', 'BirthDate', 'Gender', and 'PostalCode'. The study viewed 'MaskType' as a categorical factor that impacted therapy results. Our purpose was to improve the clustering process and support the selection of patients in need of higher intervention by considering risk factors[22]. In this phase, initial clustering formed five clusters, and the metrics reported a Silhouette Score of 0.2961

and a Davies-Bouldin Index of 0.9728. Treatments were planned based on the new and improved metrics. The significance of 'LeaksLargePercentage', 'DeviceType', and 'UsageDuration' was again proven by the ANOVA analysis. Using these features, the clustering process produced nine divisions that were well-separated and consistent. The approach to interventions and how surveys were conducted was regularly reviewed and adapted to fit the updated clusters. This part stressed the use of risk-based metrics in future phases, considering how physiological data such as blood pressure can relate to a patient's ability to stick to treatment.

D. Advanced Clustering Iteration: Integrating Intervention and Monitoring Survey Data

To further enhance personalization and extend the clustering framework, we integrated both the Intervention and Monitoring Survey datasets in the following iteration. The columns in 'Intervention' were 'ServiceId', 'JobTypeCode', 'Status', and 'ReferenceDate', while 'monitoring survey' featured 'QuestionId', 'AnswerValue', and 'ExecutionDate'. The initial Mean Shift clustering divided the datasets into five groups, with a Silhouette Score of 0.2912 and a Davies-Bouldin Index of 0.8432. This step was vital for planning how wearable biometric information would be integrated in the future [23]. Clusters were labeled and intervention strategies were applied based on their specific traits. Interventions were given to clusters showing high leakage and compliance was highlighted for clusters with low leakage. The recommendations for medical surveys were also adjusted; patients in the highest-risk clusters got checked using PSQI, SF-36, and ESS in person, while those in the moderate group were recommended to take SF-36 and BDI surveys with the help of a physician. Remote surveys (ESS, PSQI subsections) were still offered to low-risk clusters for medical screenings. They aimed to integrate body data including heart rate variations, changes in oxygen, and visual blood vessel information into improved clustering and treatment processes in later upgrades.

E. Iterative Refinements for Biomarker Integration

This process improved personalization, accuracy, and strategic success. Furthermore, during this phase, the steps were set in place to include digital biomarkers into future iterations. This method is an upgrade compared to the original K-Means approach. The aim is to use a dynamic, data-based strategy that puts patients' needs first. This framework improves personalization through continuous refining of clusters, assigning treatments on the fly, and recommending medical surveys. This progression ultimately advances the broader goal of personalized healthcare, ensuring adaptive, effective, and patient-centred adherence strategies for OSA management.

V. Conclusion

The research introduces an updated iterative clustering model for CPAP adherence optimization through its progression from K-means toward Mean Shift clustering methods, which eliminates the need for predefined cluster numbers and supports dynamic patient grouping. It links CPAP data, along with health reports, to determine the best plan for each patient based on their needs. By repeatedly clustering patients, researchers were able to fine-tune adherence patterns and respond quickly with appropriate interventions. Their plans varied in response to each occasion, making sure the riskiest groups got stronger attention, such as consistent tracking and visiting, whilst those who were less risky were helped through less invasive means from afar. By applying this approach, the benefits to patients and use of resources were well balanced. In every iteration, the medical survey was updated, as it moved in-person assessments to higher-risk patients and remote ones for low-risk groups. Introducing this process made it easier and more exact to track compliance. By assessing risk factors and using CPAP usage data, more physiological risks like leakage and inconsistent use were found, these might explain the poor long-term outcomes. While digital biomarkers like HRV, ODI, and RVO were not yet included in the launch, this study paves the way for their use in enhanced future models. CPAP adherence can now be managed more effectively by using adaptive clustering together with feedback-driven personalization. The results demonstrate the potential for future integration of physiological data streams, making this framework highly scalable for large-scale real-world deployment in OSA treatment.

VI. Result

The proposed Mean Shift iterative clustering methodology enabled dynamic pattern classification and continuous refinement of CPAP adherence groupings. They started the clustering process using standard CPAP usage results, resulting in nine distinct groups. Depending on whether a person adhered to the rules or not, the strategies selected for them changed. The analysis included risk factors, demographic data, and CPAP usage trends, creating five initial clusters after further stages. Analysis and clustering with ANOVA-based features further helped to create nine stable clusters, making them more accurate and meaningful for clinical use. The creation of intervention methods was possible due to understanding how patients respond to treatment. Patients who had repeated issues following the program were managed with more intensive monitoring and advice, while others were checked on at a distance. Patients were divided into groups based on their risk, with those in the high-risk groups receiving in-person surveys, and those in the low-risk groups undergoing monitoring over the phone or through the internet. The last development included combining intervention logs and patient-reported survey results to enhance cluster accuracy and allow real-time adjustments to treatment plans. Patient feedback results enabled the dynamic adjustment of intervention strategies and contributed to personalized therapy delivery. Following these phases, real-time biometric monitoring is now feasible, supported by the system's capability to offer scalable, data-driven CPAP adherence monitoring for future clinical implementation.

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